A VECTOR AND MELLIN–FUNCTIONAL ROUTE FROM A MERTENS WALK TO A SYMMETRIZED ξ-FAMILY VERSION 0.6: ODD LOG-PRIME MOBIUS KERNEL

Contents

1. Motivation from phase-walk geometry 1
2. The arithmetic two-channel object 2
3. Dirichlet series and the direct family 4
4. Known scales and the squarefree channel 5
5. Replacing ζ by the completed function 6
6. The symmetrized λ-family 7
7. The Mellin operator induced by the functional equation 8
8. Inverse Mellin kernel and corrected atomic interpretation 9
8.1. Corrected Mertens channel 10
8.2. Corrected squarefree channel 10
8.3. Corrected symmetrized arithmetic object 10
9. The conditional proof route 11
10. Choice of the parameters λ 12
11. The odd log-prime Mertens kernel 12
11.1. The te Riele–Odlyzko logical bridge 13
11.2. Prime-index expansion and the expected zero singularities 13
11.3. The Odlyzko–te Riele finite block for the odd kernel 14
12. What must be proved next 15
13. Notebook history 16
References 16

Remark 0.1. This notebook is exploratory in nature. It records a chain of ideas starting from numerical phase-walk pictures of the Mertens function and a squarefree correction, and ending with a proposed Mellin–functional family built from the completed zeta function. Statements labelled as lemmas or propositions are intended as formal or conditional statements. Statements labelled as heuristics, experiments, warnings, or open problems are not proofs of the Riemann hypothesis.

1. Motivation from phase-walk geometry

The initial object was not a Dirichlet series but a geometric walk. The ordinary Mertens function is

$$M(x) = \sum_{n \le x} \mu(n), \tag{1.1}$$

where μ is the Möbius function. The numerical pictures motivating this notebook considered the growth of a walk over Mertens, with an added real component coming from the squarefree counting error.

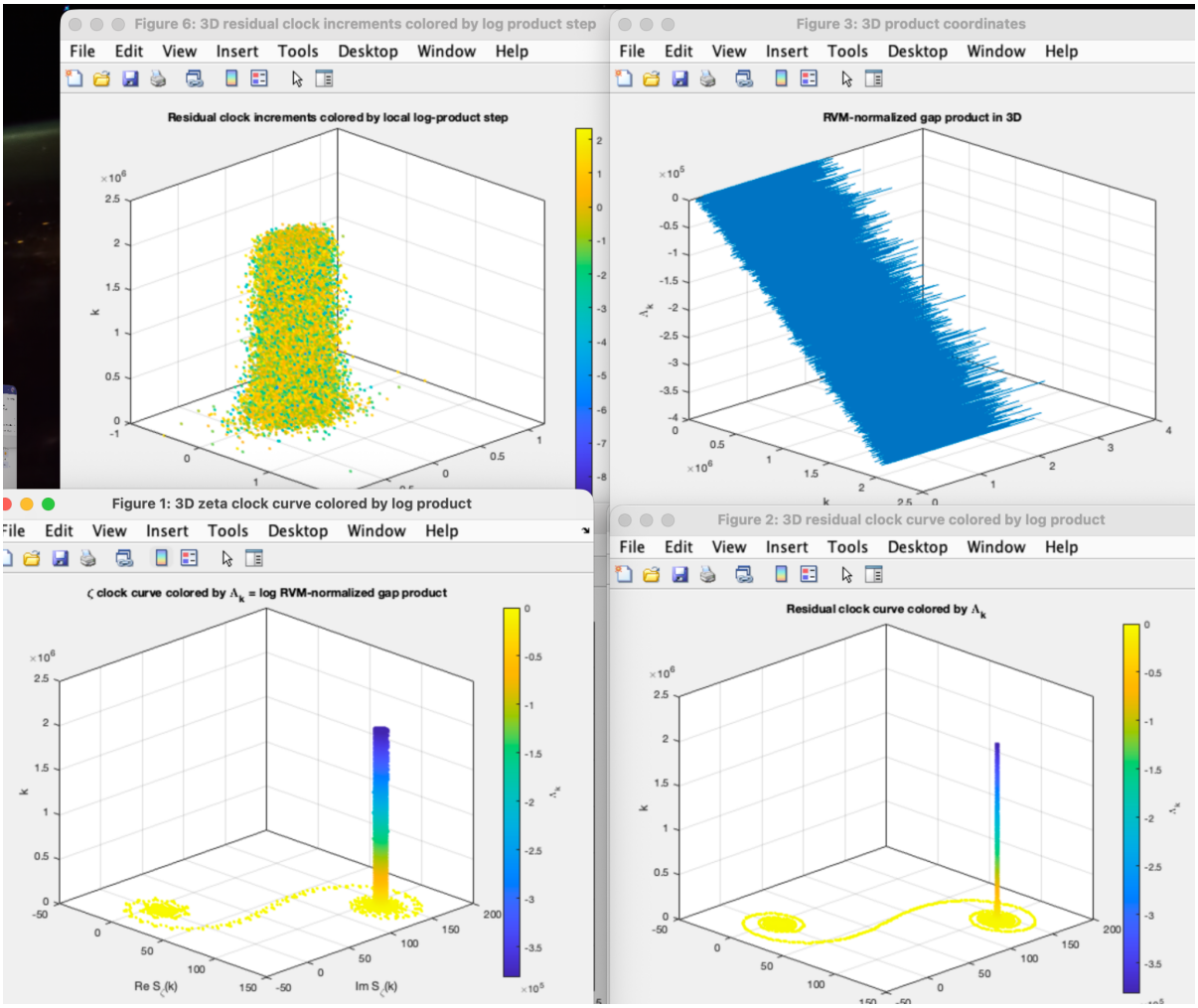


FIGURE 1. Uploaded numerical phase-walk figures. The relevant qualitative observation is the growth of a three-dimensional cloud or surface when the Mertens walk is augmented by a second arithmetic component. This figure is used only as motivation, not as evidence for a theorem.

Experiment 1.1. The computational logs used a file containing $455,052,511$ values of $\mu(n)$. In the million-by-million scan the endpoint was reported at

$$N = 455,052,511, \qquad M(N) = -5414, \qquad \theta_{\text{end}} \approx 0.251833^\circ, \tag{1.2}$$

and the fitted endpoint angle statistic was reported in the form

$$\tan\theta(N) \approx AN^b, \qquad A \approx 0.2089756943, \qquad b \approx -0.145023366. \tag{1.3}$$

These numerical values are recorded as exploratory data. They do not constitute a proof of any asymptotic statement.

2. The arithmetic two-channel object

Let

$$S_2(x) = \sum_{n \le x} \mu(n)^2 \tag{2.1}$$

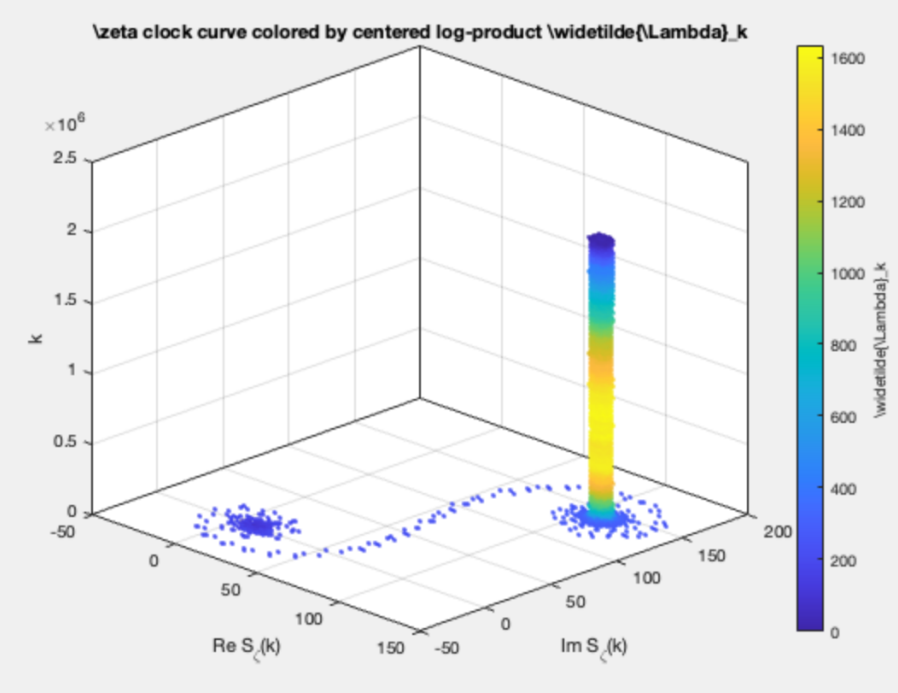


FIGURE 2. Uploaded centered/log-product three-dimensional visualization. The figure motivates an anisotropic viewpoint: the Mertens axis and the squarefree-error axis need not have the same natural scaling.

be the squarefree counting function. Define the squarefree error by

$$E_2(x) = S_2(x) - \frac{x}{\zeta(2)}. \tag{2.2}$$

The centered real component used in the geometric walk may also be written with the opposite sign,

$$R(x) = \frac{x}{\zeta(2)} - S_2(x) = -E_2(x). \tag{2.3}$$

The basic two-channel vector is therefore

$$Z_\mu(x) = R(x) + iM(x). \tag{2.4}$$

Equivalently, in terms of E_2,

$$Z_\mu(x) = -E_2(x) + iM(x). \tag{2.5}$$

Remark 2.1. The natural scales of the two components are different. The Mertens channel is associated with the critical scale $x^{1/2}$, whereas the squarefree-error channel is associated with the scale $x^{1/4}$ after the substitution $x = t^2$. This is the reason for considering $M(t)$ and $E_2(t^2)$ together.

Definition 2.2. For a complex parameter λ, define the direct combined channel by

$$C_\lambda(t) = M(t) + \lambda E_2(t^2). \tag{2.6}$$

This is the direct, unsymmetrized arithmetic channel.

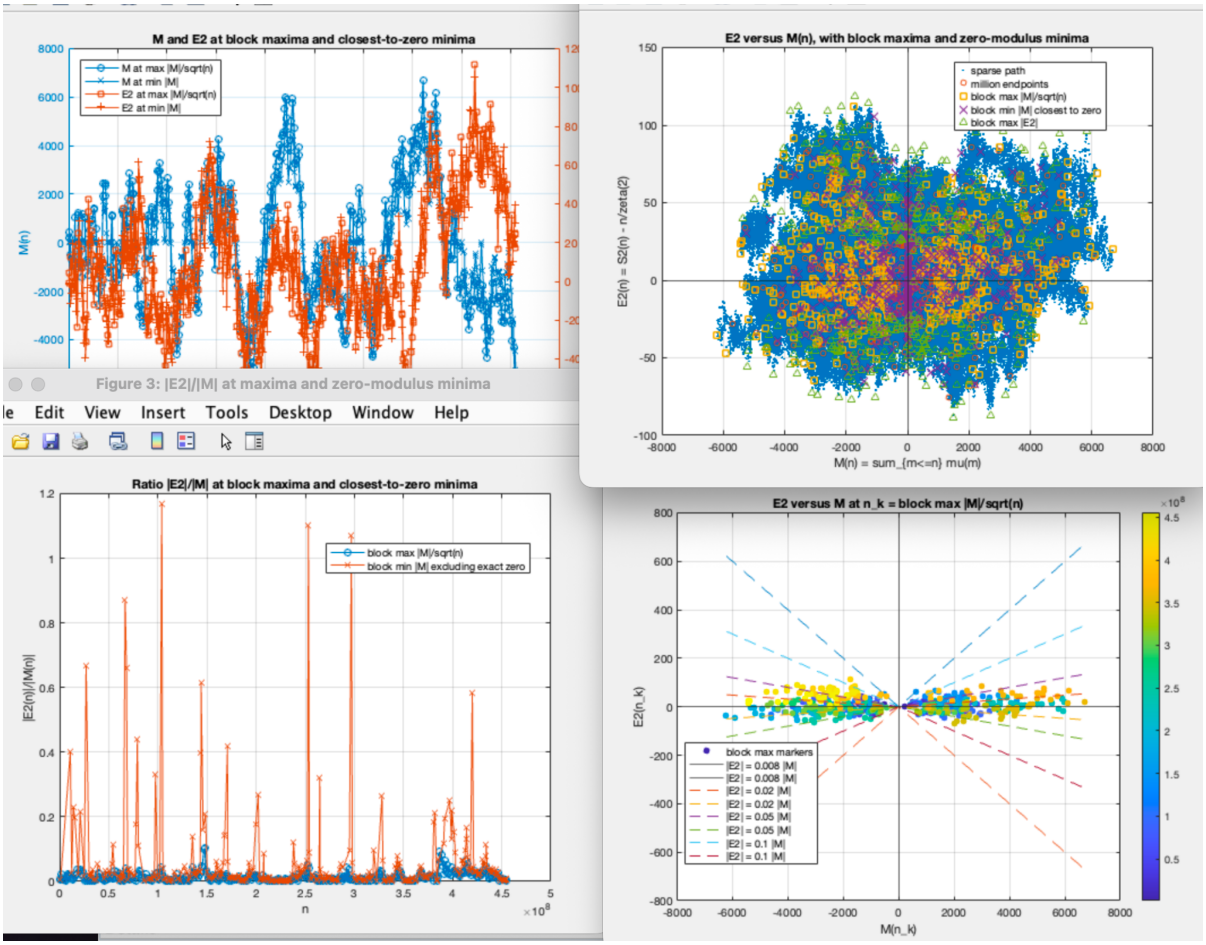


FIGURE 3. Uploaded numerical comparison panels from the Mertens and squarefree-error experiments. The panels are treated here as computational observations only.

3. Dirichlet series and the direct family

The standard Dirichlet series identities are

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}, \qquad \Re s > 1, \tag{3.1}$$

and

$$\sum_{n=1}^{\infty} \frac{\mu(n)^2}{n^s} = \frac{\zeta(s)}{\zeta(2s)}, \qquad \Re s > 1. \tag{3.2}$$

These are standard identities in analytic number theory; see [2, 4].

The re-scaled squarefree channel is produced formally by

$$\frac{\zeta(s/2)}{\zeta(s)} = \left(\sum_{a=1}^{\infty} a^{-s/2}\right)\left(\sum_{b=1}^{\infty} \frac{\mu(b)}{b^s}\right). \tag{3.3}$$

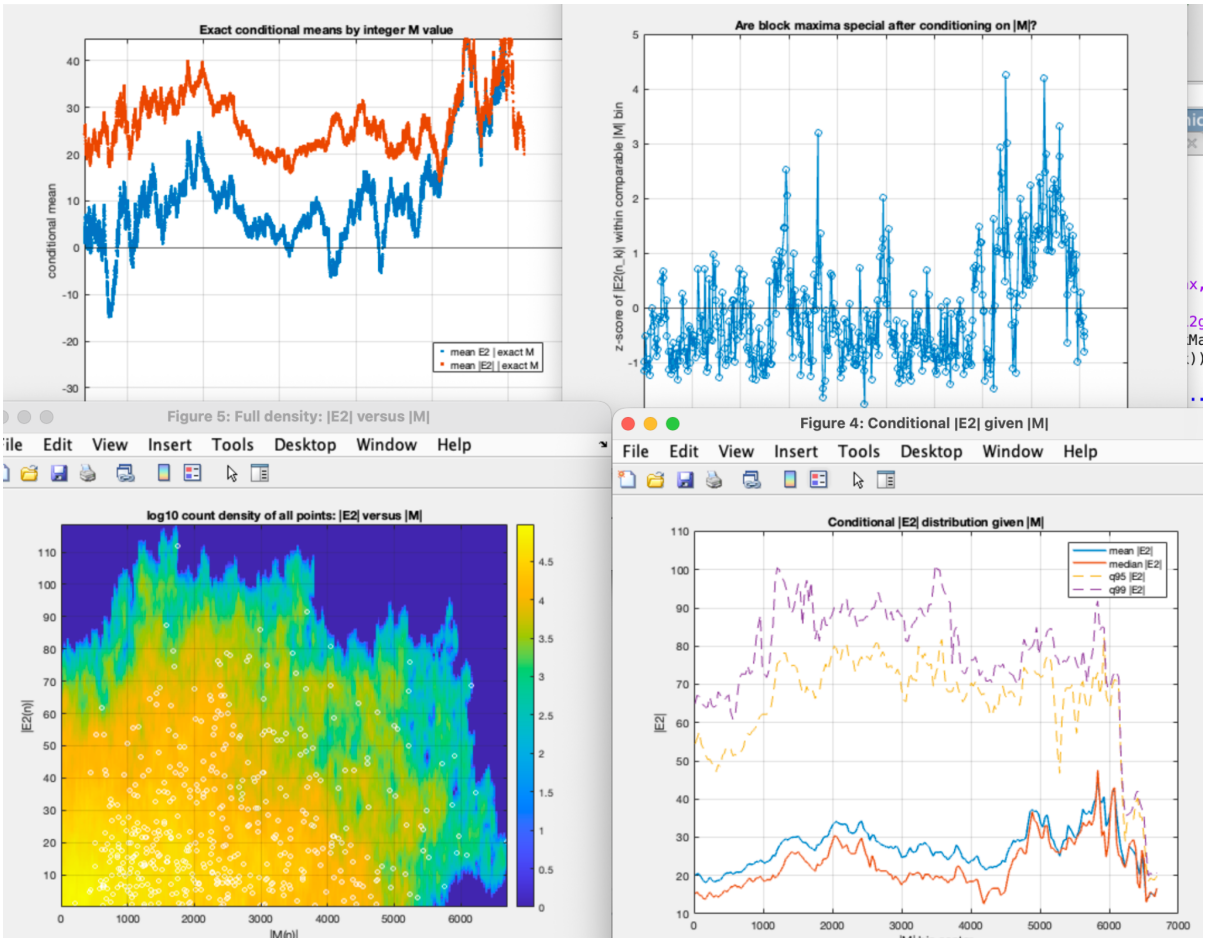


FIGURE 4. Uploaded numerical comparison panels from the later geometric investigation. These images motivate the search for a two-channel analytic formulation rather than a single Euclidean norm.

Consequently, its summatory form is

$$\sum_{a^{1/2}b\le t}\mu(b)=\sum_{b\le t}\mu(b)\left\lfloor\frac{t^2}{b^2}\right\rfloor \tag{3.4}$$

$$=S_2(t^2). \tag{3.5}$$

Thus the centered direct family corresponds formally to

$$D_\lambda(s)=\frac{1}{\zeta(s)}+\lambda\left(\frac{\zeta(s/2)}{\zeta(s)}-\frac{\zeta(s)}{\zeta(2)}\right). \tag{3.6}$$

Warning 3.1. The direct family C_λ alone is not a new analytic mechanism. Two independent values of λ recover $M(t)$ and $E_2(t^2)$ by linear algebra. The direct family is useful as a compact notation, but it is not by itself a proof strategy.

4. Known scales and the squarefree channel

The error term in squarefree counting is a standard object. Write

$$R_2(x)=S_2(x)-\frac{x}{\zeta(2)}. \tag{4.1}$$

The general k-free error is often denoted

$$R_k(x) = Q_k(x) - \frac{x}{\zeta(k)}. \tag{4.2}$$

Mossinghoff, Oliveira e Silva, and Trudgian record the expected scale

$$R_k(x) = O_\varepsilon\left(x^{1/(2k)+\varepsilon}\right), \tag{4.3}$$

and prove large oscillation results and extensive computations for $k = 2, 3, 4, 5$ [7]. For $k = 2$, the expected scale is

$$R_2(x) = O_\varepsilon\left(x^{1/4+\varepsilon}\right). \tag{4.4}$$

Remark 4.1. Equation (4.4) is recorded as the natural squarefree-error scale. This notebook does not claim a proof of (4.4). The purpose is to embed the squarefree channel into a vectorial analytic framework together with the Mertens channel.

Proposition 4.2. *Assume that both estimates*

$$M(t) = O_\varepsilon\left(t^{1/2+\varepsilon}\right) \tag{4.5}$$

and

$$E_2(t^2) = O_\varepsilon\left(t^{1/2+\varepsilon}\right) \tag{4.6}$$

hold for every $\varepsilon > 0$. Then, for every fixed λ,

$$C_\lambda(t) = O_{\varepsilon,\lambda}\left(t^{1/2+\varepsilon}\right). \tag{4.7}$$

Proof. This follows immediately from (2.6), (4.5), and (4.6). □

Warning 4.3. A single estimate of the form (4.7) for one value of λ does not recover separate estimates for M and E_2. Two independent values of λ recover the vector algebraically, but this is still a change of coordinates unless an additional analytic operation is introduced.

5. Replacing ζ by the completed function

Define

$$A(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \tag{5.1}$$

and

$$\xi(s) = A(s)\zeta(s). \tag{5.2}$$

The completed zeta function satisfies the functional equation

$$\xi(s) = \xi(1-s), \tag{5.3}$$

which is standard; see [2, 3, 4].

Lemma 5.1. *Away from the explicitly known zeros and poles of the factor $A(s)$, passing from $\zeta(s)$ to $\xi(s)$ is reversible.*

Proof. Equation (5.2) gives

$$\zeta(s) = \frac{\xi(s)}{A(s)}. \tag{5.4}$$

Similarly,

$$\frac{1}{\zeta(s)} = \frac{A(s)}{\xi(s)}. \tag{5.5}$$

Thus no information about the nontrivial zeros is lost by using ξ, except for the already known factors in $A(s)$. □

The two non-reflected components become

$$\frac{1}{\zeta(s)} = \frac{A(s)}{\xi(s)}, \tag{5.6}$$

and

$$\frac{\zeta(s/2)}{\zeta(s)} = \frac{A(s)\zeta(s/2)}{\xi(s)}. \tag{5.7}$$

Using (5.3), the reflected components are

$$\frac{A(1-s)}{\xi(s)}, \tag{5.8}$$

and

$$\frac{A(1-s)\zeta((1-s)/2)}{\xi(s)}. \tag{5.9}$$

Definition 5.2. The completed vector family is

$$\mathcal{V}(s) = \frac{1}{\xi(s)} \begin{pmatrix} A(s) \\ A(s)\zeta(s/2) \\ A(1-s) \\ A(1-s)\zeta((1-s)/2) \end{pmatrix}. \tag{5.10}$$

Remark 5.3. The vector $\mathcal{V}(s)$ is not a projectivization. Its first component is exactly $1/\zeta(s)$ by (5.6). Therefore a bound on the full vector would control the Mertens channel. This is different from taking a quotient of two components, which would remove the scalar denominator and lose the size information.

6. The symmetrized λ-family

For $\lambda \in \mathbb{C}$, define the numerators

$$N_\lambda^\pm(s) = A(s)(1+\lambda\zeta(s/2)) \pm A(1-s)(1+\lambda\zeta((1-s)/2)). \tag{6.1}$$

The symmetrized functions are

$$F_\lambda^\pm(s) = \frac{N_\lambda^\pm(s)}{\xi(s)}. \tag{6.2}$$

Equivalently,

$$F_\lambda^+(s) + F_\lambda^-(s) = 2\frac{A(s)(1+\lambda\zeta(s/2))}{\xi(s)}, \tag{6.3}$$

$$F_\lambda^+(s) - F_\lambda^-(s) = 2\frac{A(1-s)(1+\lambda\zeta((1-s)/2))}{\xi(s)}. \tag{6.4}$$

Thus the pair $(F_\lambda^+, F_\lambda^-)$ contains both the original and reflected channels.

Definition 6.1. Set

$$P^\pm(s) = A(s) \pm A(1-s), \tag{6.5}$$

and

$$Q^\pm(s) = A(s)\zeta(s/2) \pm A(1-s)\zeta((1-s)/2). \tag{6.6}$$

Then

$$N_\lambda^\pm(s) = P^\pm(s) + \lambda Q^\pm(s). \tag{6.7}$$

Lemma 6.2. *If*

$$N_1^{\pm}(s) = 0 \qquad \text{and} \qquad N_i^{\pm}(s) = 0, \tag{6.8}$$

then

$$P^{\pm}(s) = 0 \qquad \text{and} \qquad Q^{\pm}(s) = 0. \tag{6.9}$$

Proof. Using (6.7), the assumptions (6.8) are

$$P^{\pm}(s) + Q^{\pm}(s) = 0, \tag{6.10}$$

and

$$P^{\pm}(s) + iQ^{\pm}(s) = 0. \tag{6.11}$$

Subtracting (6.10) from (6.11) gives

$$(i-1)Q^{\pm}(s) = 0. \tag{6.12}$$

Since $i - 1 \neq 0$, this gives $Q^{\pm}(s) = 0$. Substitution into (6.10) gives $P^{\pm}(s) = 0$. □

Warning 6.3. Lemma 6.2 is only an algebraic non-cancellation test. It does not prove that $F_1^{\pm}$ or $F_i^{\pm}$ is holomorphic or bounded in any half-plane. The analytic work must come from a Mellin construction of these functions from sums, not from assuming that the quotient (6.2) has no poles.

7. The Mellin operator induced by the functional equation

The quotient of the two completion factors is

$$\chi(s) = \frac{A(1-s)}{A(s)} = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \tag{7.1}$$

This is the standard factor in the zeta functional equation

$$\zeta(s) = \chi(s)\zeta(1-s). \tag{7.2}$$

Definition 7.1. Let $\mathcal{K}$ denote the formal Mellin multiplier with symbol $\chi(s)$. For a function or distribution f with Mellin transform $\mathcal{M}f$, define

$$\mathcal{M}[\mathcal{K}f](s) = \chi(s)\mathcal{M}[f](s). \tag{7.3}$$

Equivalently, where the inverse Mellin transform is justified,

$$(\mathcal{K}f)(x) = \int_0^{\infty} K(xy)f(y)\frac{dy}{y}, \tag{7.4}$$

with kernel

$$K(u) = \frac{1}{2\pi i}\int_{(c)} \chi(s)u^{-s}\,ds. \tag{7.5}$$

Warning 7.2. Definition 7.1 is formal unless a function space, a contour, and convergence or distributional interpretation are specified. This is one of the main analytic gaps in the proposed route.

Heuristic 7.3. This is a heuristic argument. If C_λ is the direct summatory object in (2.6), then the completed functional symmetrization suggests a pair of the form

$$\mathcal{C}_\lambda^{\pm}(t) = C_\lambda(t) \pm (\mathcal{K}C_\lambda)(t). \tag{7.6}$$

The direct part corresponds to $D_\lambda(s)$. The transformed part corresponds to multiplication by $\chi(s)$, which is the analytic shadow of the reflection $s \mapsto 1-s$. The precise atomic version, including the corrected squarefree jumps at $\sqrt{m}$, is given in Section 8.

8. Inverse Mellin kernel and corrected atomic interpretation

This section corrects a possible ambiguity in the previous formulation. The Mellin transform belongs naturally to the atomic measure associated with a summatory function, not directly to the summatory step function itself. This distinction changes the correct interpretation of the squarefree channel $E_2(x^2)$.

Using the equivalent form of the zeta functional equation factor,

$$\chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s), \tag{8.1}$$

one has, initially in the usual strip of convergence and then by analytic continuation,

$$\chi(s) = 2\int_0^\infty \cos(2\pi u) u^{-s}\, du. \tag{8.2}$$

Equivalently, the inverse Mellin kernel is the reciprocal cosine kernel

$$K(u) = \frac{2}{u}\cos\left(\frac{2\pi}{u}\right) \tag{8.3}$$

in the distributional sense compatible with (7.5).

Proposition 8.1. *Let*

$$dA = \sum_{j=1}^{J} a_j \delta_{r_j}, \qquad r_j > 0, \tag{8.4}$$

be a finite atomic measure, and set

$$D_A(s) = \int_0^\infty x^{-s}\, dA(x) = \sum_{j=1}^{J} a_j r_j^{-s}. \tag{8.5}$$

Then multiplication by $\chi(s)$ sends dA to the oscillatory density

$$d(\mathcal{K}A)(y) = 2\sum_{j=1}^{J} \frac{a_j}{r_j} \cos\left(\frac{2\pi y}{r_j}\right) dy, \tag{8.6}$$

and the corresponding reflected summatory function is

$$(\mathcal{K}A)(x) = \frac{1}{\pi}\sum_{j=1}^{J} a_j \sin\left(\frac{2\pi x}{r_j}\right). \tag{8.7}$$

Proof. For finite sums the computation is direct. From (8.2),

$$\chi(s)D_A(s) = 2\sum_{j=1}^{J} a_j r_j^{-s} \int_0^\infty \cos(2\pi u) u^{-s}\, du \tag{8.8}$$

$$= 2\sum_{j=1}^{J} \frac{a_j}{r_j} \int_0^\infty \cos\left(\frac{2\pi y}{r_j}\right) y^{-s}\, dy, \tag{8.9}$$

where $y = r_j u$. This proves (8.6). Integrating (8.6) from 0 to x gives (8.7). □

Warning 8.2. For infinite arithmetic measures, equations (8.6) and (8.7) must be interpreted with a regularization or in a distributional sense. A convenient regularized version is

$$(\mathcal{K}_\eta A)(x) = \frac{1}{\pi}\sum_{j} a_j \sin\left(\frac{2\pi x}{r_j}\right) e^{-\eta r_j}, \qquad \eta > 0, \tag{8.10}$$

followed, if possible, by a limit $\eta \downarrow 0$.

8.1. **Corrected Mertens channel.** For the Mertens channel, the atomic data are

$$r_n = n, \qquad a_n = \mu(n). \tag{8.11}$$

Therefore the formal reflected channel is

$$(\mathcal{K}M)(x) = \frac{1}{\pi} \sum_{n\geq 1} \mu(n) \sin\left(\frac{2\pi x}{n}\right), \tag{8.12}$$

again subject to regularization as in (8.10). The symmetrized Mertens component is thus

$$M(x) \pm (\mathcal{K}M)(x) = \sum_{n\geq 1} \mu(n) \left(\mathbf{1}_{n\leq x} \pm \frac{1}{\pi} \sin\left(\frac{2\pi x}{n}\right)\right). \tag{8.13}$$

8.2. **Corrected squarefree channel.** For the re-scaled squarefree channel the variable is x, while the summatory function is $S_2(x^2)$. Hence

$$S_2(x^2) = \sum_{m\leq x^2} \mu(m)^2 = \sum_{\sqrt{m}\leq x} \mu(m)^2. \tag{8.14}$$

Thus the jumps are not located at m in the x-variable. They are located at

$$r_m = \sqrt{m}, \qquad a_m = \mu(m)^2. \tag{8.15}$$

Consequently,

$$(\mathcal{K}S_2(\cdot^2))(x) = \frac{1}{\pi} \sum_{m\geq 1} \mu(m)^2 \sin\left(\frac{2\pi x}{\sqrt{m}}\right), \tag{8.16}$$

with the same regularization caveat.

The centered discrete version is

$$\widetilde{E}_2(x^2) = \sum_{\sqrt{m}\leq x} \left(\mu(m)^2 - \frac{1}{\zeta(2)}\right). \tag{8.17}$$

It differs from the continuous-centering convention (2.2) by a bounded term:

$$\widetilde{E}_2(x^2) - E_2(x^2) = \frac{x^2 - \lfloor x^2 \rfloor}{\zeta(2)}. \tag{8.18}$$

The corrected reflected centered squarefree channel is therefore

$$(\mathcal{K}\widetilde{E}_2)(x) = \frac{1}{\pi} \sum_{m\geq 1} \left(\mu(m)^2 - \frac{1}{\zeta(2)}\right) \sin\left(\frac{2\pi x}{\sqrt{m}}\right), \tag{8.19}$$

again interpreted through (8.10) or another justified summability method.

Warning 8.3. The earlier double-sum notation based on points $b\sqrt{a}$ is useful for deriving $\zeta(s/2)/\zeta(s)$, but it is not the clean atomic description of the summatory function $S_2(x^2)$. Once the summatory function is written as (8.14), the correct jumps in the x-variable are at $\sqrt{m}$, with weights $\mu(m)^2$. The centered version uses weights $\mu(m)^2 - 1/\zeta(2)$.

8.3. **Corrected symmetrized arithmetic object.** Using the discrete centering (8.17), define

$$\widetilde{C}_\lambda(x) = M(x) + \lambda \widetilde{E}_2(x^2). \tag{8.20}$$

The corrected Mellin-reflected combination is

$$\widetilde{\mathcal{C}}^{\pm}_\lambda(x) = M(x) + \lambda \widetilde{E}_2(x^2) \pm \frac{1}{\pi} \sum_{n\geq 1} \mu(n) \sin\left(\frac{2\pi x}{n}\right) \tag{8.21}$$

$$\pm \frac{\lambda}{\pi} \sum_{m\geq 1} \left(\mu(m)^2 - \frac{1}{\zeta(2)}\right) \sin\left(\frac{2\pi x}{\sqrt{m}}\right). \tag{8.22}$$

The four objects to be studied are

$$\widetilde{\mathcal{C}}_1^+, \qquad \widetilde{\mathcal{C}}_i^+, \qquad \widetilde{\mathcal{C}}_1^-, \qquad \widetilde{\mathcal{C}}_i^-. \tag{8.23}$$

Remark 8.4. The analytic crash point is now sharper. One must prove cancellation in (8.22) without first proving the RH-scale bounds for $M(x)$ or $E_2(x^2)$. In particular, the squarefree part is not governed by jumps at integers in the x-variable, but by jumps at the square-root lattice $\{\sqrt{m} : m \geq 1\}$.

9. The conditional proof route

The target is not to prove estimates by listing zeros. The target is to construct the symmetrized family from sums and prove analytic continuation or bounds in a half-plane.

Conjecture 9.1. *For $\lambda = 1$ and $\lambda = i$, the Mellin transforms attached to the symmetrized sums $\mathcal{C}_\lambda^+$ and $\mathcal{C}_\lambda^-$ admit holomorphic continuation to*

$$\Re s > \frac{1}{2}, \tag{9.1}$$

with growth bounds sufficient for the usual Perron or inverse Mellin shifts.

Conjecture 9.2. *For $\Re s > 1/2$, a zero of $\xi(s)$ cannot satisfy simultaneously*

$$P^\pm(s) = 0 \qquad \text{and} \qquad Q^\pm(s) = 0 \tag{9.2}$$

for both choices of sign in the manner needed to hide all poles of the family $F_1^\pm, F_i^\pm$.

Proposition 9.3. *Assume Conjecture 9.1 and Conjecture 9.2. Then $\xi(s)$ has no zeros in the half-plane $\Re s > 1/2$. Consequently, using the functional equation* (5.3)*, the nontrivial zeros of $\zeta(s)$ lie on the critical line.*

Proof. Suppose that s_0 satisfies

$$\Re s_0 > \frac{1}{2} \qquad \text{and} \qquad \xi(s_0) = 0. \tag{9.3}$$

By Conjecture 9.1, the functions represented by the symmetrized sums for $\lambda = 1$ and $\lambda = i$ have no pole at s_0. In the quotient representation (6.2), any pole from $\xi(s_0) = 0$ must therefore be cancelled by the corresponding numerator.

For a fixed sign, cancellation for both $\lambda = 1$ and $\lambda = i$ gives

$$N_1^\pm(s_0) = 0 \qquad \text{and} \qquad N_i^\pm(s_0) = 0. \tag{9.4}$$

By Lemma 6.2, this implies

$$P^\pm(s_0) = 0 \qquad \text{and} \qquad Q^\pm(s_0) = 0. \tag{9.5}$$

This is excluded by Conjecture 9.2. Hence no such s_0 exists in the half-plane $\Re s > 1/2$.

The functional equation (5.3) reflects zeros across the critical line. The absence of zeros in $\Re s > 1/2$ therefore gives the corresponding absence in $\Re s < 1/2$ for nontrivial zeros, and the remaining nontrivial zeros lie on $\Re s = 1/2$. □

Warning 9.4. Proposition 9.3 is conditional. It does not prove the Riemann hypothesis. The unsolved part is Conjecture 9.1, together with the non-cancellation statement in Conjecture 9.2.

10. Choice of the parameters λ

The pair

$$\Lambda = \{1, i\} \tag{10.1}$$

is the smallest complex pair that separates the two numerator components $P^{\pm}$ and $Q^{\pm}$ through Lemma 6.2. The alternative real pair

$$\Lambda_{\mathbb{R}} = \{1, -1\} \tag{10.2}$$

also separates directions, but the complex pair is less aligned with possible real-axis cancellations.

A larger test family is

$$\Lambda_4 = \{1, -1, i, -i\}. \tag{10.3}$$

This adds redundancy but does not alter the algebraic content. The real requirement is a uniform analytic estimate for the associated symmetrized family, not a numerical search over λ.

Remark 10.1. Testing random points or known zeros computationally can detect accidental degeneracies of the numerators $N_\lambda^{\pm}$, but it cannot prove Conjecture 9.1. The required proof is a half-plane estimate or continuation theorem for the Mellin-transformed symmetrized sums.

11. The odd log-prime Mertens kernel

This section records the new kernel suggested by the numerical suppression experiments. Unlike the two-channel vector in Sections 2–10, the new object is a one-channel odd projection of a Möbius sum twisted by the logarithm of the indexed prime.

Definition 11.1 (Odd log-prime kernel)**.** Fix

$$c = \frac{\pi}{2}, \qquad t \in \mathbb{R}. \tag{11.1}$$

Let p_k denote the k-th prime. Define

$$Y_t(X) = \sum_{k \le X} \mu(k) \sin\big(t \log(c p_k)\big), \tag{11.2}$$

and the associated complex kernel

$$A_t(X) = \sum_{k \le X} \mu(k) (c p_k)^{it}. \tag{11.3}$$

Then

$$Y_t(X) = \Im A_t(X). \tag{11.4}$$

Equivalently,

$$Y_t(X) = \frac{1}{2i} \left(c^{it} \sum_{k \le X} \mu(k) p_k^{it} - c^{-it} \sum_{k \le X} \mu(k) p_k^{-it} \right). \tag{11.5}$$

The numerical test motivating this kernel was the comparison of

$$\frac{Y_t(K)}{\sqrt{K}} \tag{11.6}$$

against sign-permutation nulls preserving the zero set $\mu(k) = 0$. For $t = \pi/2$, the observed tail ratios for high quantiles and maxima were substantially smaller than one. This motivates the following analytic question.

Question 11.2 (Critical-scale objective)**.** Can one prove, without first assuming a zero-free half-plane equivalent to the desired conclusion, that for fixed $t \neq 0$

$$Y_t(X) = O_{t,\varepsilon}\left(X^{1/2+\varepsilon}\right) \qquad (X \to \infty) \tag{11.7}$$

for every $\varepsilon > 0$, or at least a sharp $O_t(\sqrt{X})$-scale estimate for a smoothed version of Y_t?

11.1. **The te Riele–Odlyzko logical bridge.** For the ordinary Mertens function, the classical bridge is direct:

$$M(X) = O_\varepsilon(X^{1/2+\varepsilon}) \quad \Longrightarrow \quad \frac{1}{\zeta(s)} \text{ is holomorphic in } \Re s > \frac{1}{2}. \tag{11.8}$$

Thus a critical-scale Mertens estimate rules out zeros of ζ in $\Re s > 1/2$. Odlyzko and te Riele then work with the explicit formula under the resulting critical-line zero structure and use lattice reduction to find large values contradicting the Mertens conjecture [6].

The corresponding Dirichlet series for the odd log-prime kernel is

$$D_t(s) = \sum_{k=1}^{\infty} \frac{\mu(k)\sin(t\log(cp_k))}{k^s}. \tag{11.9}$$

The exact analogue of the first step is the following elementary Abel implication.

Lemma 11.3 (Abel implication for the odd kernel)**.** *Assume that for every $\varepsilon > 0$*

$$Y_t(X) = O_{t,\varepsilon}(X^{1/2+\varepsilon}). \tag{11.10}$$

Then $D_t(s)$ is holomorphic in every half-plane

$$\Re s > \frac{1}{2} + \varepsilon. \tag{11.11}$$

Equivalently, the critical-scale estimate forces holomorphic continuation of D_t to the open half-plane $\Re s > 1/2$, up to the usual ε-formulation.

Proof. For $\Re s > 1$, summation by parts gives

$$D_t(s) = s\int_1^{\infty} Y_t(x)x^{-s-1}\,dx. \tag{11.12}$$

If $Y_t(x) = O_{t,\varepsilon}(x^{1/2+\varepsilon})$, the integral converges locally uniformly for $\Re s > 1/2 + \varepsilon$. This gives holomorphy there. □

Therefore, to copy the te Riele–Odlyzko bridge, one must show that $D_t(s)$ inherits non-removable singularities from the zeros of ζ. The next calculation identifies the expected singularities.

11.2. **Prime-index expansion and the expected zero singularities.** The prime number theorem for the indexed prime gives the asymptotic expansion

$$p_k = k\log k\left(1 + \frac{\log\log k - 1}{\log k} + O\left(\frac{(\log\log k)^2}{(\log k)^2}\right)\right). \tag{11.13}$$

Consequently, for fixed t,

$$p_k^{it} = k^{it}(\log k)^{it}\left(1 + it\frac{\log\log k - 1}{\log k} + O_t\left(\frac{(\log\log k)^2}{(\log k)^2}\right)\right). \tag{11.14}$$

The leading model for D_t is therefore

$$D_t^{(0)}(s) = \frac{c^{it}}{2i}\sum_{k=1}^{\infty}\mu(k)(\log k)^{it}k^{-(s-it)} - \frac{c^{-it}}{2i}\sum_{k=1}^{\infty}\mu(k)(\log k)^{-it}k^{-(s+it)}. \tag{11.15}$$

Formally, since

$$\sum_{k=1}^{\infty} \mu(k)k^{-z} = \frac{1}{\zeta(z)}, \tag{11.16}$$

we may write

$$D_t^{(0)}(s) = \frac{c^{it}}{2i}(-\partial_z)^{it}\frac{1}{\zeta(z)}\Big|_{z=s-it} - \frac{c^{-it}}{2i}(-\partial_z)^{-it}\frac{1}{\zeta(z)}\Big|_{z=s+it}. \tag{11.17}$$

This identity is to be understood at this stage as a Mellin-symbol calculation: the factor $(\log k)^{it}$ corresponds to an imaginary-order differentiation multiplier.

Let ρ be a simple zero of ζ. Near $z = \rho$,

$$\frac{1}{\zeta(z)} \sim \frac{1}{\zeta'(\rho)(z-\rho)}. \tag{11.18}$$

The imaginary-order derivative of the model pole gives

$$(-\partial_z)^{it}\frac{1}{z-\rho} \sim \Gamma(1+it)(z-\rho)^{-1-it}, \tag{11.19}$$

and similarly for $-it$. Hence the leading model predicts singularities of D_t at

$$s = \rho + it \qquad \text{and} \qquad s = \rho - it. \tag{11.20}$$

Their leading coefficients are

$$C_+(\rho,t) = \frac{c^{it}\Gamma(1+it)}{2i\,\zeta'(\rho)}, \qquad C_-(\rho,t) = -\frac{c^{-it}\Gamma(1-it)}{2i\,\zeta'(\rho)}. \tag{11.21}$$

For simple zeros and fixed real $t \neq 0$, these coefficients are nonzero in the leading model.

Proposition 11.4 (Formal te Riele-type implication for the odd kernel)**.** *Assume that the exact Dirichlet series $D_t(s)$ has non-removable singularities at $s = \rho \pm it$ whenever ρ is a zero of ζ, as predicted by the leading model* (11.17)–(11.21)*. Then*

$$Y_t(X) = O_{t,\varepsilon}(X^{1/2+\varepsilon}) \quad \textit{for every } \varepsilon > 0 \tag{11.22}$$

implies that $\zeta(s)$ has no zeros in $\Re s > 1/2$.

Proof. If $Y_t(X) = O_{t,\varepsilon}(X^{1/2+\varepsilon})$, Lemma 11.3 makes $D_t(s)$ holomorphic in $\Re s > 1/2$. If ζ had a zero $\rho = \beta + i\gamma$ with $\beta > 1/2$, then the assumed singularity inheritance would give a singularity of D_t at $s = \rho + it$ or $s = \rho - it$. Both points have real part $\beta > 1/2$, contradicting holomorphy. □

Remark 11.5 (The exact gap). The missing non-formal step is not the lattice computation. It is the proof that the exact series $D_t(s)$, with the true indexed primes p_k, has the non-removable singularities predicted by (11.21). If the correction terms in (11.14) cancel those singularities, the bridge fails. If they do not cancel them, the same logical first step used in the Mertens setting is available.

11.3. **The Odlyzko–te Riele finite block for the odd kernel.** Once the singularity-inheritance step is established, the lattice stage is not used to prove the zero-free half-plane. It is used to work with the explicit formula under the critical-line structure forced by a putative critical-scale bound, exactly as in the Mertens disproof strategy.

Let $u = \log X$, and let h be a smooth kernel in the u-variable with rapidly decaying Fourier transform $\widehat{h}$. The normalized smoothed function is

$$\mathcal{Y}_{t,h}(u) = \int_{\mathbb{R}} e^{-v/2} Y_t(e^v) h(u-v)\,dv. \tag{11.23}$$

The expected finite block has the form

$$P_{t,T}(u) = \Im \sum_{0<\gamma_n\le T} \left(\alpha_n(t)\widehat{h}(\gamma_n+t)e^{i(\gamma_n+t)u} + \beta_n(t)\widehat{h}(\gamma_n-t)e^{i(\gamma_n-t)u}\right), \tag{11.24}$$

where, in the leading model,

$$\alpha_n(t) \asymp \frac{c^{it}\Gamma(1+it)}{\zeta'(1/2+i\gamma_n)}, \qquad \beta_n(t) \asymp -\frac{c^{-it}\Gamma(1-it)}{\zeta'(1/2+i\gamma_n)}. \tag{11.25}$$

The lattice problem is the inhomogeneous phase alignment problem

$$(\gamma_n \pm t)u + \arg(\alpha_n \text{ or } \beta_n) \approx \frac{\pi}{2} \pmod{2\pi}, \tag{11.26}$$

with weights $|\alpha_n\widehat{h}(\gamma_n+t)|$ and $|\beta_n\widehat{h}(\gamma_n-t)|$. This is the direct analogue of the Odlyzko–te Riele finite block, but for the odd log-prime kernel rather than for the ordinary Mertens function.

Warning 11.6 (What the lattice step can and cannot close). The lattice step can test or certify the finite smoothed block $P_{t,T}$. It does not by itself prove the abrupt estimate (11.7). The non-circular bridge consists of: (i) singularity inheritance for D_t; (ii) critical-scale bound implying holomorphy of D_t; (iii) the Odlyzko–te Riele finite-block analysis under the resulting critical-line structure. The exact singularity-inheritance statement is the next mathematical point to prove.

12. What must be proved next

The route has four nontrivial requirements.

Question 12.1. For the odd log-prime series $D_t(s)$ in (11.9), can one prove the non-removable singularity inheritance predicted in (11.20)–(11.21)? Equivalently, does every simple zero ρ of ζ produce non-cancelled singularities of D_t at $s = \rho \pm it$?

Question 12.2. Can one define a function space or distribution space in which $\mathcal{K}$, with multiplier $\chi(s)$, acts on the arithmetic summatory functions associated with C_λ?

Question 12.3. Can one prove Conjecture 9.1 directly from the corrected symmetrized sums $\widetilde{\mathcal{C}}_\lambda^{\pm}$ in (8.22), without using any statement equivalent to the absence of zeros of $\zeta(s)$ in $\Re s > 1/2$?

Question 12.4. Can one prove a non-circular cancellation estimate for

$$M(x) \pm \frac{1}{\pi}\sum_{n\ge1}\mu(n)\sin\left(\frac{2\pi x}{n}\right) \tag{12.1}$$

and simultaneously for the square-root squarefree analogue

$$\widetilde{E}_2(x^2) \pm \frac{1}{\pi}\sum_{m\ge1}\left(\mu(m)^2 - \frac{1}{\zeta(2)}\right)\sin\left(\frac{2\pi x}{\sqrt{m}}\right), \tag{12.2}$$

without first proving RH-scale estimates for the unsymmetrized summatory functions?

Question 12.5. Can one prove the no-cancellation condition (9.2) using known properties of $A(s)$, $\zeta(s/2)$, and $\zeta((1-s)/2)$?

Remark 12.6. The route may fail if proving Conjecture 9.1 requires a bound on $1/\zeta(s)$ in $\Re s > 1/2$. In that case the argument would become circular. The only non-circular possibility is to obtain continuation or estimates from the symmetrized Mellin operator itself.

13. Notebook history

Version 0.1 records the geometric starting point: a Mertens walk augmented by the squarefree-error direction.

Version 0.2 records the direct two-channel family $C_\lambda = M + \lambda E_2(t^2)$, together with the warning that a finite family of λ's is only a change of coordinates.

Version 0.3 records the completed ξ-vector and the symmetrized functions $F_\lambda^\pm$.

Version 0.4 isolates the real analytic gap: the Mellin operator $\mathcal{K}$ and the need to prove holomorphic continuation or growth bounds for the symmetrized sums without assuming the desired zero-free region.

Version 0.5 corrects the atomic interpretation of the Mellin-reflected squarefree channel. For $E_2(x^2)$, the jumps in the x-variable occur at $\sqrt{m}$, with weights $\mu(m)^2$ or $\mu(m)^2 - 1/\zeta(2)$ after discrete centering. This supersedes the earlier informal double-sum presentation for the reflected summatory object.

Version 0.6 adds the odd log-prime Möbius kernel $Y_t(X) = \sum_{k \le X} \mu(k) \sin(t \log(cp_k))$, records the Abel-continuation bridge for its Dirichlet series, derives the leading zero-singularity model at $s = \rho \pm it$, and identifies the exact remaining gap: proving non-removable singularity inheritance for the true indexed-prime series.

References

[1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie, 1859.

[2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, second edition, revised by D. R. Heath-Brown, Oxford University Press, 1986. ISBN: 9780198533696.

[3] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974. ISBN: 9780486417400.

[4] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, 2004. ISBN: 9780821836330.

[5] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in *Analytic Number Theory*, Proceedings of Symposia in Pure Mathematics, vol. 24, American Mathematical Society, 1973, pp. 181–193. DOI: 10.1090/pspum/024/0337821.

[6] A. M. Odlyzko and H. J. J. te Riele, *Disproof of the Mertens conjecture*, Journal für die reine und angewandte Mathematik, vol. 357, 1985, pp. 138–160. DOI: 10.1515/crll.1985.357.138.

[7] M. J. Mossinghoff, T. Oliveira e Silva, and T. S. Trudgian, *The distribution of k-free numbers*, Mathematics of Computation, vol. 90, 2021, pp. 907–929. DOI: 10.1090/mcom/3581. arXiv:1912.04972.

[8] D. Zagier, *The Mellin transform and other useful analytic techniques*, in *Quantum Field Theory I: Basics in Mathematics and Physics*, Springer, 2006, pp. 305–323.